

3. SEQUENCE OPERATIONS AND SIGNAL ANALYSIS

Experiment 3.1

Program:

```
% Addition and Multiplication of Two Sequences
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Sample index
```

```
n = 0:10;
```

```
% Sequence 1
```

```
x1 = [1 2 3 4 5 6 7 8 9 10 11];
```

```
% Sequence 2
```

```
x2 = [11 10 9 8 7 6 5 4 3 2 1];
```

```
% Addition of sequences
```

```
y_add = x1 + x2;
```

```
% Multiplication of sequences
```

```
y_mul = x1 .* x2;
```

```
% Plotting
```

```
figure;
```

```
subplot(2,2,1);
```

```
stem(n,x1,'filled');
```

```
grid on;
```

```
title('Sequence x_1(n)');
```

```
xlabel('n');
```

```
ylabel('Amplitude');
```

```
subplot(2,2,2);
```

```
stem(n,x2,'filled');
```

```
grid on;
```

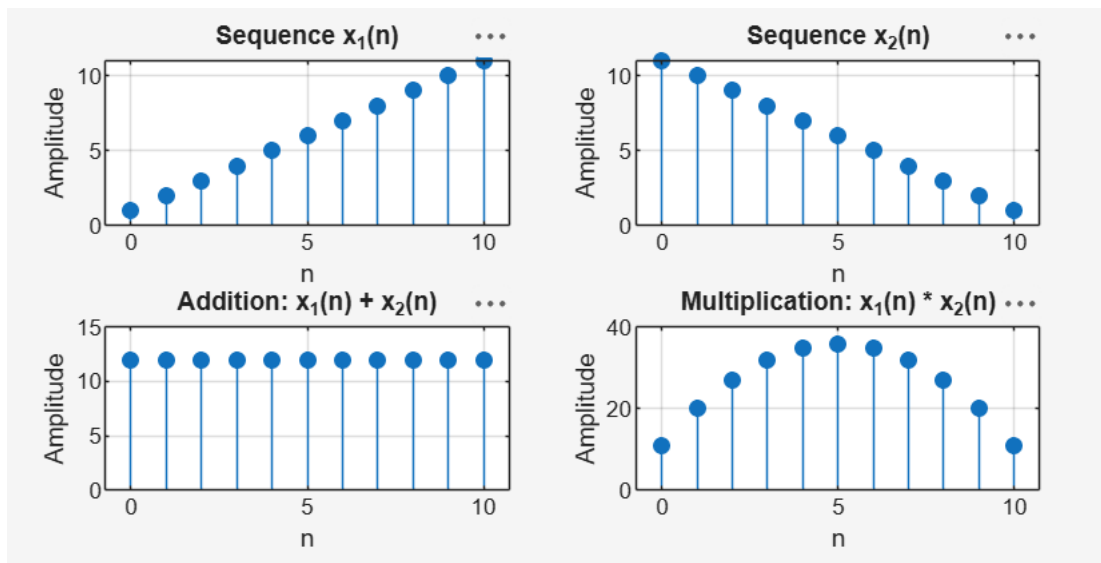
```
title('Sequence x_2(n)');
```

```
xlabel('n');
```

```

ylabel('Amplitude');
subplot(2,2,3);
stem(n,y_add,'filled');
grid on;
title('Addition:  $x_1(n) + x_2(n)$ ');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,4);
stem(n,y_mul,'filled');
grid on;
title('Multiplication:  $x_1(n) * x_2(n)$ ');
xlabel('n');
ylabel('Amplitude');

```



Experiment 3.2

Energy and Power Calculation of Given Signals

Program (MATLAB)

```
% Energy and Power Calculation of Signals

clc;

clear;

close all;

% Time index
n = -10:10;

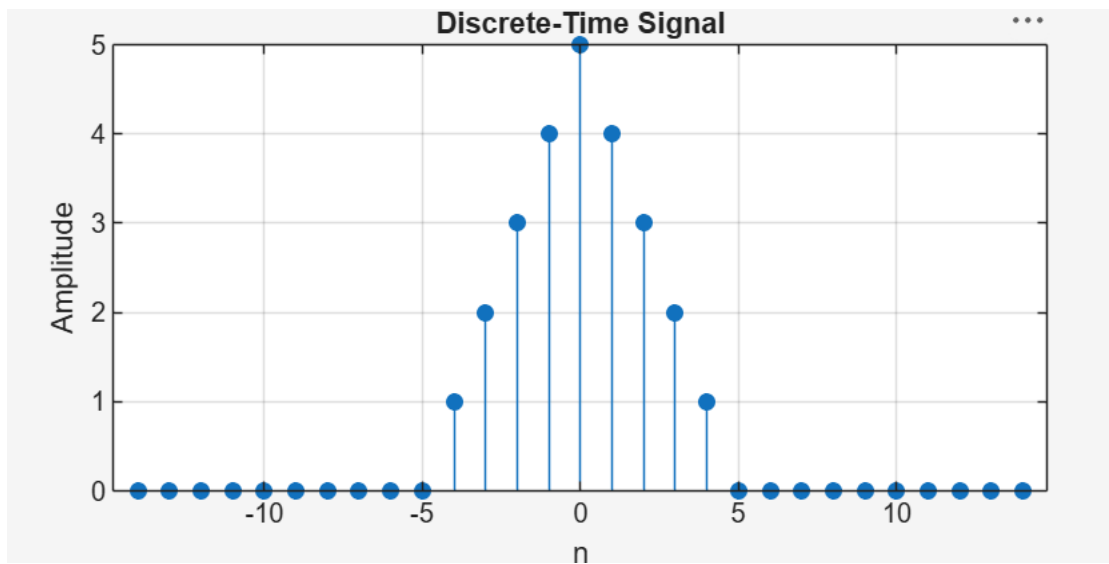
% Signal
x = [zeros(1,10) 1 2 3 4 5 4 3 2 1 zeros(1,10)];

% Energy Calculation
E = sum(abs(x).^2);

% Power Calculation
P = E/length(x);

% Display Results
disp(['Energy of the signal = ', num2str(E)]);
disp(['Power of the signal = ', num2str(P)]);

% Plot Signal
figure;
stem(n,x,'filled');
grid on;
title('Discrete-Time Signal');
xlabel('n');
ylabel('Amplitude');
```



Command Window

```
Energy of the signal = 85  
Power of the signal = 2.931  
>>
```